How Income Perceptions and Living Standards Shape Inequality and New Product Diffusion

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- Two-part survey study based on a UK-representative sample with $n = 875$.
- Model data combining:
  - novel spline-based density and distribution estimation (DDFS);
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Contributions:

1. Assess inequality using univariate/multivariate Lorenz curves and Gini coef.
2. Explore the implications of perceived income on consumer behaviour and product diffusion.

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2. Purchasing decisions depend on how income is expressed through observable consumption and lifestyle: **standard of living**.
3. Since true income is often unobserved, firms and market participants form *perceptions of income* based on visible indicators of *standard of living*, such as postcode, house value, car value, or expenditure patterns.

Estimating Actual Income, X, and Standard of Living, Z

Notation:
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Survey Part I

- Collect data on actual income-X and standard of living-Z;
- Let Z be a positive random variable defined as

$$Z = w_1 C_1 + \ldots + w_k C_k, \tag{1}$$

where $0 < w_i < 1$, $\sum_{i=1}^{k} w_i = 1$ are weights, reflecting the importance of the corresponding standard of living component, C_i, $i = 1, ..., k$.

- Obtain DDFS estimate of the univariate (marginal) distribution F_X and F_Z.

Estimating Perceived Income, $\tilde{X}$

Survey Part II

- Collect data on perceived income ($\tilde{X}$):
  (i) Each returning participant is reminded about their income, X, and informed about their standard of living, Z.

Welcome back to the second part of the **Consumer Expenditure and Possession Survey!**

You completed the first part of the survey 168 days ago (on 2024-07-19), where you were asked to indicate the range of your annual income, given the following possible ranges:

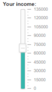
1. £0-15,000
2. £15,001-30,000
3. £30,001-45,000
4. £45,001-60,000
5. £60,001-75,000
6. £75,001-90,000
7. £90,001-105,000
8. £105,001-120,000
9. £120,001 or Above

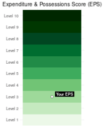
You said that your **annual income** falls within the 5th income range, namely £60,001-75,000 (also referred to hereby as £60k-75k).

Recall that "you" refers to *all members of your family*.

You also answered some questions related to your income and possessions (e.g. the value of your house/rent, the value of your vehicle, and how much you spend on groceries, holidays and leisure). Based on your responses, we have calculated an **Expenditure & Possessions Score (EPS)** measure, which ranges from Level 1 (lowest EPS) to Level 10 (highest EPS). Specifically, your EPS value is 2.3161, which corresponds to **Level 3 of the EPS scale.**

To summarize, your reported income and your EPS are depicted below.

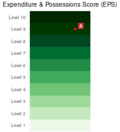




(ii) Then, they are asked to estimate the likelihood of a fellow participant having a particular annual income given their standard of living level, z_i.

Consider another participant from Part I of this survey. Let's call this person Participant A. Participant A provided responses about their individual income, expenditure, and possessions.

Based on their individual responses, Participant A has an **Expenditure & Possessions Score (EPS)**, that falls in **Level 9**.



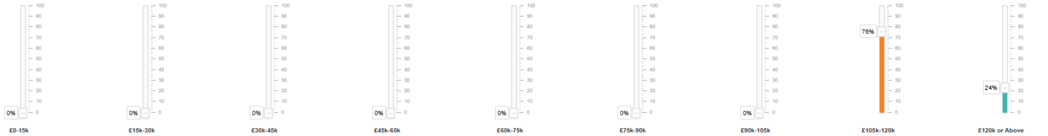
Given Participant A's EPS level (i.e., Level 9), what is their individual annual income?

Below, please **estimate the *likelihood* (in %)** that Participant A's annual income falls within the different income ranges.

Use the sliders provided below to set the probability (in %) that Participant A's annual income falls into each income range. The total of your allocated probabilities should sum up to 100% (please note that, if the total exceeds 100% at any point, the program will recalculate the probabilities so that the total is below 100%, but the relative proportions allocated to the different income ranges remain the same).

Probability Left to Allocate:

0 %



- Build the ecdf of perceived income $\tilde{X}$
  - (i) Conditional ecdf for a given standard of living level z_i

$$\tilde{F}_{i,n}(x|z_i) = \frac{1}{n}\sum_{j=1}^{n} \tilde{F}_j(x|z_i), \quad i = 1, \ldots, m = 10. \tag{2}$$

where $\tilde{F}_j(x|z_i)$ is the income distribution perceived by the j-th individual for an anonymous person with an hypothetical z_i, visible for j.

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  - (ii) And then we obtain the following discrete estimate of $F_{\tilde{X}}(x)$:

$$\tilde{F}_n(x) = \sum_{i=1}^{m}\tilde{F}_{i,n}(x|z_i)\hat{f}_Z(z_i), \tag{3}$$

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- Obtain DDFS estimate of $F_{\tilde{X}}(x)$, by fitting $\tilde{F}_n(x)$.

Joint (Bivariate) Modelling

Two important pairs:

- **Actual Income X & Standard of Living Z**
- **Perceived Income $\tilde{X}$ & Standard of Living Z**

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Following Sklar's theorem, the joint cdf of X-Z, $\tilde{X}$-Z can be written as

$$F_{X,Z}(x,z) = C(F_X(x), F_Z(z)),$$
$$F_{\tilde{X},Z}(x,z) = \tilde{C}(F_{\tilde{X}}(x), F_Z(z)),$$

where marginals, $F_X(x)$, $F_Z(z)$, $F_{\tilde{X}}(x)$, and copulas, C, $\tilde{C}$, are estimated separately through DDFS + 2-dimensional Gaussian copula, C_ρ^{Ga}.

Inequality Measurement

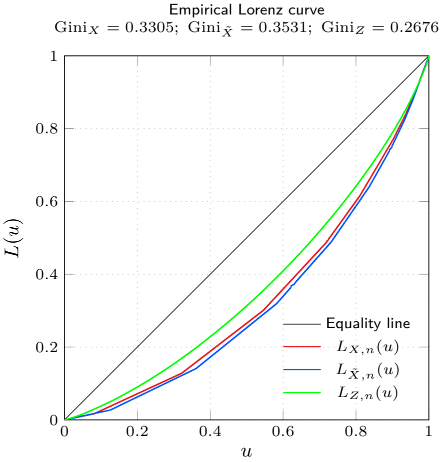
- *Marginal (Univariate) Inequality*:

$$L_X(u) = \frac{1}{\mu}\int_0^u F_X^{-1}(\nu)d\nu;$$
$$L_{\tilde{X}}(u) = \frac{1}{\mu}\int_0^u F_{\tilde{X}}^{-1}(\nu)d\nu;$$
$$L_Z(u) = \frac{1}{\mu}\int_0^u F_Z^{-1}(\nu)d\nu.$$

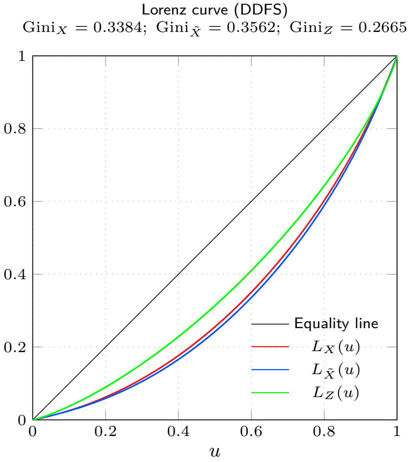
- *Joint (bivariate) inequality* (Grothe et al., 2022):

$$\mathbf{L}_{C,L_X^{-1},L_Z^{-1}}(u_X, u_Z) = C(L_X^{-1}(u_X), L_Z^{-1}(u_Z));$$
$$\mathbf{L}_{C,L_{\tilde{X}}^{-1},L_Z^{-1}}(u_{\tilde{X}}, u_Z) = \tilde{C}(L_{\tilde{X}}^{-1}(u_{\tilde{X}}), L_Z^{-1}(u_Z)).$$

Univariate Lorenz Curves

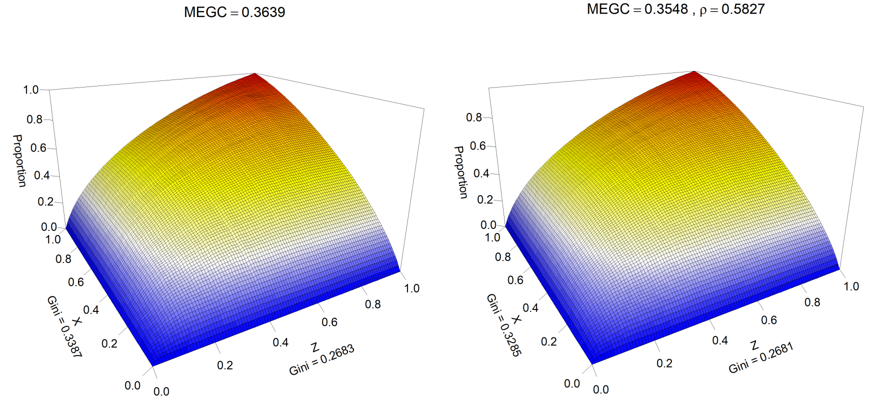


(a) $L_n(u)$



(b) DDFS on $L(u)$

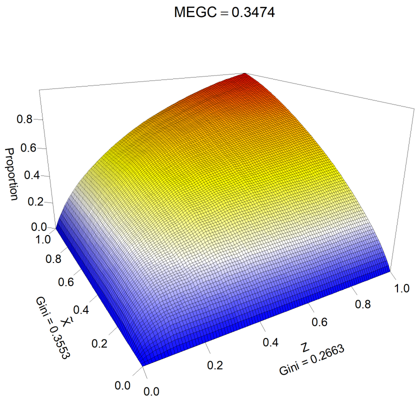
X-Z Lorenz Curves



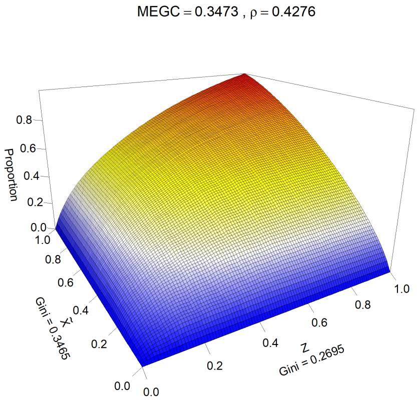
(a) $(x_k, z_k | x_k)$, $k = 1, \ldots, n$

(b) $\hat{C}^{\text{Ga}}_{\hat{\rho}}(\hat{F}_X(x), \hat{F}_Z(z))$

$\tilde{X}$-Z Lorenz Curves

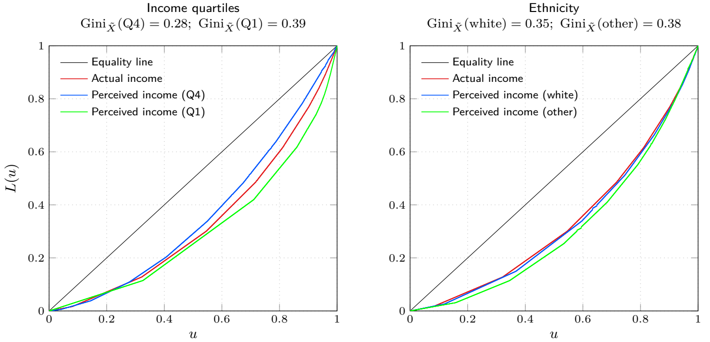


(a) $(\tilde{x}_k | z_k, z_k)$, $k = 1, \ldots, v$



(b) $\hat{C}^{\text{Ga}}_{\hat{\rho}}(\hat{F}_{\tilde{X}}(x), \hat{F}_Z(z))$

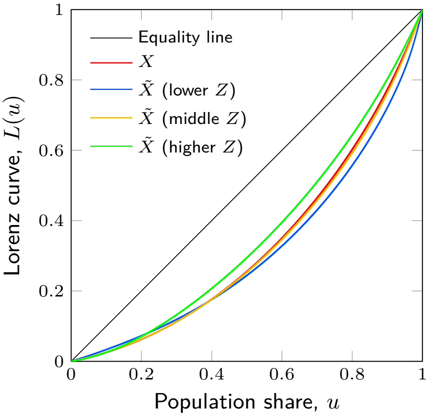
Cross-Sectional Inequality Perception



(a) Income quartiles　　(b) Ethnicity

Figure 4: Actual income and perceived income Lorenz curves.

Perceived Inequality Depends on the Inference Technology



- Different **observer groups** generate different perceived income distributions.
- Firms infer **purchasing power** from observable indicators of standard of living (perceived income distribution).

New Product Diffusion

- New product diffusion assumes a new product is a type of innovation and models the time to (first) purchase $\tilde{T}$ in a population over time.
- **Bass, 1969**:
  - seminal quantitative framework for new product diffusion;
  - behavioural rational in terms of imitative and innovative behaviour (*social contagion*).
- Similar diffusion patterns may also result from heterogeneity in adoption thresholds across consumers (Van den Bulte and Stremersch, 2004): $\longrightarrow$ **Income Threshold Model**.

New Product Diffusion

Threshold model (Van den Bulte and Stremersch, 2004): a purchase occurs when the new product price $p(t)$ drops below the reservation price Θ, where $\Theta = \alpha\tilde{X} + \beta$.

Proposition 1. *We have*

$$\{\tilde{T} \le t\} = \left\{ \bigcup_{0 \le s \le t} \{\Theta > p(s)\} \right\} = \left\{ \Theta > \min_{0 \le s \le t} p(s) \right\}. \quad (4)$$

Diffusion curve:

$$F_{\tilde{T}}(t) = 1 - F_{\Theta}\left(\min_{0 \le s \le t} p(s) \right) = 1 - F_{\tilde{X}}\left(\min_{0 \le s \le t} [(p(s) - \beta)/\alpha] \right). \quad (5)$$

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Proposition 2. *We have*

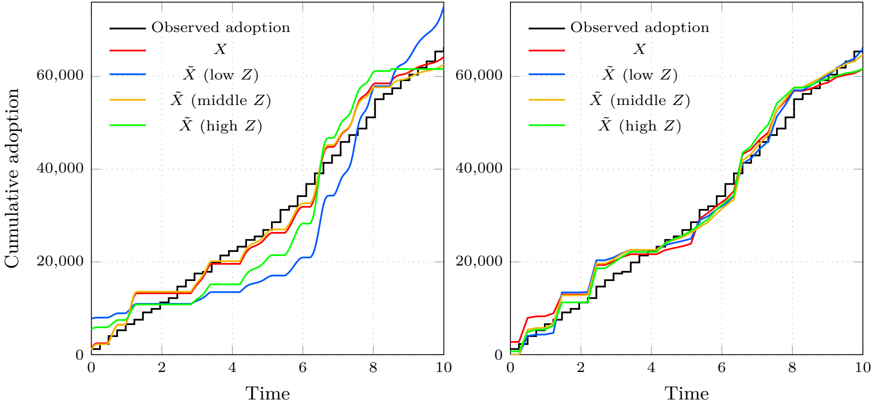
$$\{\tilde{T} \leq t\} = \bigcup_{0 \leq s \leq t} \{\Theta > p(s), Z > z(s)\} = \bigcup_{0 \leq s \leq t} \left\{\tilde{X} > (p(s) - \beta)/\alpha, Z > z(s)\right\}. \tag{6}$$

Diffusion curve:

$$F_{\tilde{T}}(t) = \iint_{\tilde{\mathcal{R}}(t)} f_{\tilde{X},Z}(\tilde{x}, z)\tilde{x}dz = \iint_{\tilde{\mathcal{R}}(t)} \tilde{c}(F_{\tilde{X}}(y), F_Z(z))f_{\tilde{X}}(\tilde{x})f_Z(z)d\tilde{x}dz \tag{7}$$

where $\tilde{\mathcal{R}}(t) = \bigcup_{0 \leq s \leq t} \{\tilde{x} > (p(s) - \beta)/\alpha, z > z(s)\}$.

Real Data Example: Nissan Leaf ($p(t)$) + Disposable Income ($z(t)$)



(a) Threshold model (5) (b) Extended threshold model (7)

Model	Threshold Model (5)	Extended Threshold Model (7)
X	2,783.69	2,650.63
$\tilde{X}$ (low Z)	7,817.49	2,463.41
$\tilde{X}$ (middle Z)	2,610.44	2,485.02
$\tilde{X}$ (high Z)	5,367.97	2,473.31

Table 1: **RMSE** for the fitted cumulative Nissan Leaf adoption paths.

$\longrightarrow$ Framework that links diffusion to pricing, affordability thresholds, perceived purchasing power, living standards, and changing economic conditions.

Discussion & Further Research

- For pricing and diffusion decisions, firms often operate with inferred purchasing power, constructed from observable standard-of-living signals.
- Different **inference technologies** yield different perceived income distributions.
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Further research:

- Specify stochastic processes (e.g., Ornstein–Uhlenbeck) for $p(t)$ and $z(t)$, enabling simulation-based forecasting of adoption under uncertainty.
- Incorporate product-specific characteristics as additional drivers of adoption.
- Perceived-income inference for diffusion takeoff, price-responsiveness, and optimal pricing.

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